Formal Lab Report: Sample A

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11/ 28 / 2023

CHEM 2081 Section 003

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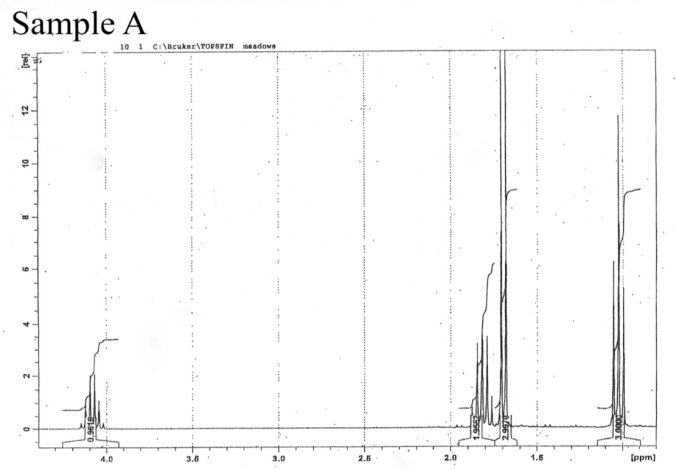
Introduction:

The purpose of this assignment was to identify an unknown chemical compound using three of the most common ways; Nuclear Magnetic Resonance (NMR), Gas Chromatography and Mass Spectrum (GC-MS), and Fourier Transform Infrared (FTIR).

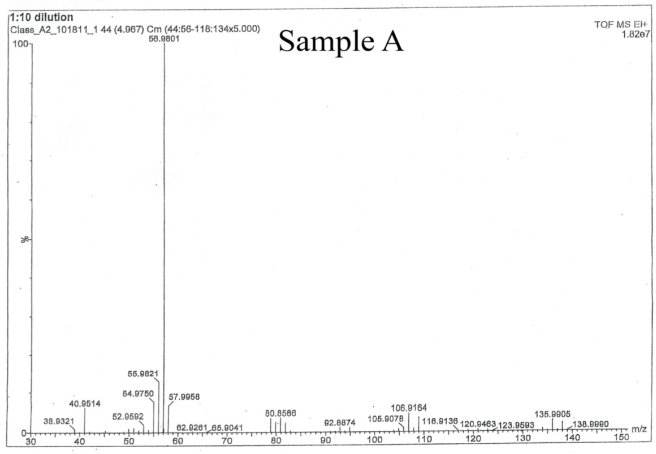
Procedure:

The analyses listed above are only three of many analyses used to identify a chemical substance. First, the Nuclear Magnetic Resonance (NMR) can determine the structure and placement of hydrogens and functional groups by using a magnet. NMR is mostly used for describing structures in organic chemistry and can specifically pinpoint all of the functional groups and hydrogens. Knowing these details of the molecule is critical and helps to understand the chemical structure as a whole. Gas Chromatography (GC) is a separation technique used for smaller molecules and it's combined with the Mass Spectrum (MS) which can be used to quantify analyses and identify unknown peaks. Specifically, when a magnetic field is applied the charged particles deflect and focus on the detector according to their mass. Lastly, the Fourier Transform Infrared (FTIR) uses beams of light and shoots them at the unknown and it reads how each beam is reflected. When sample molecules are exposed to infrared radiation, there is a change in dipole movement which causes vibrational energy. This energy is what determines the frequency of the absorption peak and the amount of absorption peaks are in response to the vibrational freedom.

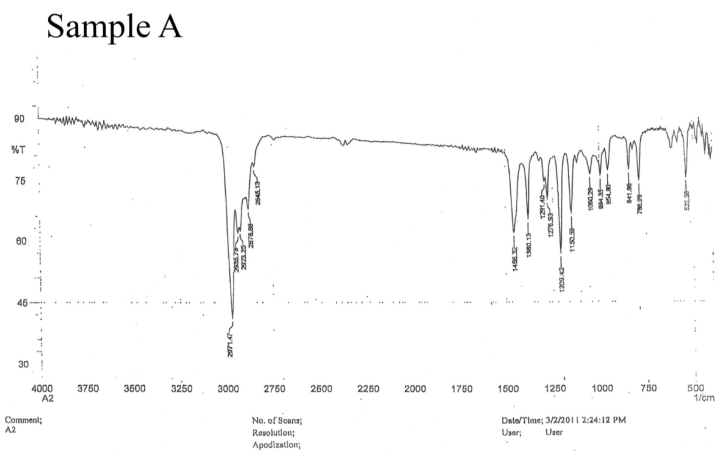
NMR Sample A:



GC-MS Sample A:



FTIR Sample A:



Conclusion:

With the data presented for the three graphs above, the unknown sample A is 2-bromobutane. First, the NMR indicates the presence of four individual peaks which indicates that this unknown sample's basic structure is a 4-carbon chain. The GC-MS indicated that the mass of M+ was 138 g/mol and that the base peak at *m/z* 56.9, M-79, indicates a loss of bromine. The FTIR shows three significant peaks. The largest peak has a frequency of 2971.74 1/cm and the second peak has a frequency of 1380.13 1/cm, which both indicate an alkane. The third peak, however, has a frequency of 532.38 1/cm, which indicates a halogen (specifically bromine). Using this data, the unknown sample was determined to be 2-bromobutane.

References:

H McNeil. Reduction of a Ketone. *Chem 2081 Organic Chemistry II Lab Manual.* Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 33-38).

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